

Nexus Document Platform 4.2 — System Architecture

Overview

Nexus Document Platform is an enterprise document-processing platform used to ingest, classify, enrich, and publish business documents. The platform consists of a WPF desktop client, Metadata Service, Document Processing Service, REST API Gateway, PostgreSQL database, object storage, and an asynchronous message broker.

Request Flow

A WPF client submits a document through the API Gateway. The gateway authenticates the request and forwards it to Document Processing Service. Processing creates a document record, stores the binary payload in object storage, and publishes a `MetadataRequested` message. Metadata Service consumes the message, extracts metadata, and writes the result back to PostgreSQL. Failed operations are sent to the retry queue.

Operational Boundaries

The WPF client must not connect directly to PostgreSQL. Database access is restricted to backend services. Object storage is used for binary content, while PostgreSQL stores metadata, processing state, and audit records.

Related Components

See `metadata-service.md`, `document-processing.txt`, `authentication.md`, and `deployment.yaml`.